

MadeUP — Integrated FP&A Analytics Project

Complete 0% → 100% Portfolio Manual

Microsoft Excel + SQL + Power BI

1. PROJECT PURPOSE

This project exists for one primary reason:

To provide credible, visible, hands-on proof that I can use Microsoft Excel, SQL, and Power BI to perform practical FP&A analysis.

This is **not** intended to prove that I have worked for a real company.

It is a **synthetic but authentic FP&A simulation**.

The company is fictional:

MadeUP

The company name is intentionally simple, like Apple, Google, Meta, Samsung, etc.

The company is a fictional global technology company whose business model, financial structure, planning processes, and reporting environment are designed to resemble a realistic modern technology organization.

The data is synthetic.

The **financial logic is realistic**.

2. WHAT THE FINISHED PROJECT MUST PROVE

At 100% completion, the project should demonstrate that I can:

Finance / FP&A

- Understand a technology company's business model
- Work with revenue and cost drivers
- Build a financial planning model

- Build an annual budget
- Analyze actual performance
- Perform Budget vs Actual analysis
- Calculate and interpret financial variances
- Build a rolling forecast
- Perform scenario analysis
- Analyze financial KPIs
- Identify risks and opportunities
- Make management recommendations

Excel

- Structure a professional financial model
- Use formulas to build financial logic
- Use `SUMIFS`
- Use `XLOOKUP`
- Use `IF` / `IFS`
- Use `IFERROR`
- Use `SUMPRODUCT` where appropriate
- Use Excel Tables
- Use PivotTables
- Use charts
- Use slicers where useful
- Build a P&L
- Build a Budget vs Actual model
- Build a forecast
- Build scenarios
- Build model checks/reconciliation

SQL

- Understand relational databases
- Create a database
- Create tables
- Define primary/foreign-key relationships
- Load synthetic financial data
- Query financial data
- Use `SELECT`
- Use `WHERE`
- Use `GROUP BY`
- Use `ORDER BY`
- Use aggregate functions
- Use `CASE`
- Use `JOIN`
- Use CTEs
- Use date logic
- Calculate financial metrics
- Analyze trends

- Identify variances
- Validate financial data
- Export analytical results

Power BI

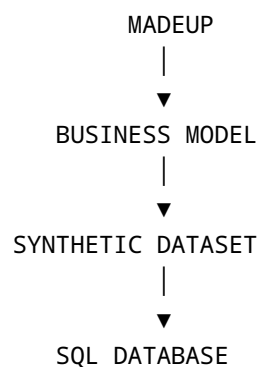
- Import financial data
- Understand data modeling
- Build relationships
- Use Power Query for basic shaping where appropriate
- Create DAX measures
- Create financial KPIs
- Build management dashboards
- Build variance visuals
- Build trend analysis
- Add interactive filtering
- Present financial insights clearly

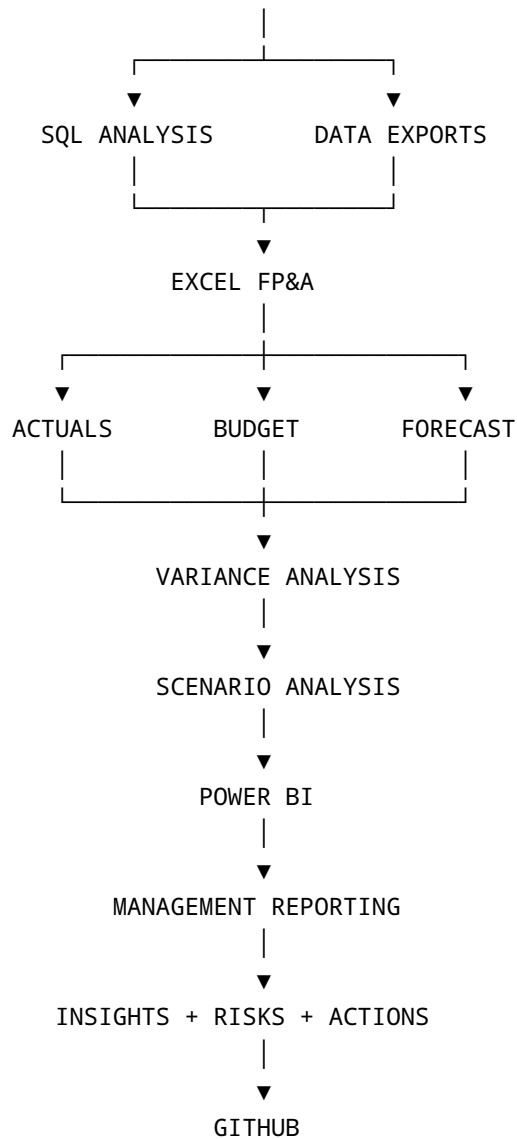
Portfolio

- Maintain a professional GitHub repository
- Document assumptions
- Document methodology
- Document the data model
- Document the financial model
- Document SQL analysis
- Document Power BI reporting
- Explain limitations
- Present management insights
- Explain the entire project in an interview

3. THE PROJECT IN ONE PICTURE

The entire project should follow this architecture:





This diagram is the project.

Every task in this manual should fit somewhere inside it.

4. WHY WE ARE USING SYNTHETIC DATA

Using synthetic data is intentional.

Public company data creates unnecessary complications because companies generally do not publish:

- internal budgets
- detailed monthly forecasts

- detailed GL transactions
- departmental budgets
- internal management reports
- employee-level financial data
- detailed product-level planning assumptions

Our objective is not to reproduce a real company's confidential FP&A environment.

Our objective is to demonstrate **how FP&A works**.

Synthetic data gives us complete control over:

- revenue
- costs
- products
- regions
- departments
- actuals
- budgets
- forecasts
- variances
- scenarios
- financial exceptions

Therefore:

The numbers are fictional, but the financial process is authentic.

The README must clearly disclose this.

5. MADEUP — THE BUSINESS

5.1 Company

MadeUP

5.2 Industry

Technology

5.3 Business profile

MadeUP is a fictional global technology company operating across:

1. Devices
2. Cloud & Software
3. Digital Services

It sells products and services to consumers and businesses across multiple geographic markets.

MadeUP is intentionally designed to have characteristics commonly found in large technology companies without copying any specific company's actual financials.

6. MADEUP BUSINESS UNITS

Business Unit 1 — Devices

Products:

- MadeUP Phone
- MadeUP Phone Pro
- MadeUP Pad
- MadeUP Book
- MadeUP Watch

Revenue driver:

$\text{Units Sold} \times \text{Average Selling Price}$

Business Unit 2 — Cloud & Software

Products:

- MadeUP Cloud
- MadeUP Workspace

Revenue driver:

$\text{Active Subscribers} \times \text{Average Revenue per Subscriber}$

Business Unit 3 — Digital Services

Products:

- MadeUP Music
- MadeUP Video
- MadeUP Store

Revenue driver:

Subscribers / Transactions × Average Revenue per User or Transaction

7. PRODUCTS

Create the following product master table.

Product ID	Product	Business Unit	Revenue Model	Base Price	Approx. COGS %
P001	MadeUP Phone	Devices	Unit Sales	\$699	62%
P002	MadeUP Phone Pro	Devices	Unit Sales	\$999	58%
P003	MadeUP Pad	Devices	Unit Sales	\$549	60%
P004	MadeUP Book	Devices	Unit Sales	\$1,199	55%
P005	MadeUP Watch	Devices	Unit Sales	\$299	50%
P006	MadeUP Cloud	Cloud & Software	Subscription	\$39	25%
P007	MadeUP Workspace	Cloud & Software	Subscription	\$24	20%
P008	MadeUP Music	Digital Services	Subscription	\$12	30%
P009	MadeUP Video	Digital Services	Subscription	\$15	35%
P010	MadeUP Store	Digital Services	Transaction	\$8	15%

These are **project assumptions**, not market data.

8. GEOGRAPHIC MARKETS

Use four regions.

Region ID	Region	Modeling Factor
R01	North America	1.00
R02	Europe	0.85
R03	Asia Pacific	0.95
R04	Latin America	0.55

The factor is simply a modeling mechanism to create different revenue scales.

It is not intended to represent actual market share.

9. DEPARTMENTS

Use eight departments.

Department ID	Department
D01	Sales
D02	Marketing
D03	Research & Development
D04	General & Administrative
D05	Information Technology
D06	Customer Operations
D07	Human Resources
D08	Facilities

These departments will drive operating expenses and headcount.

10. EXPENSE CATEGORIES

Use four categories:

Account ID	Account
A01	Payroll
A02	Operating Programs
A03	Technology & Software
A04	Facilities & Travel

This is enough detail for our first version.

Do not create 50 expense categories.

That would add complexity without adding meaningful portfolio value.

11. FINANCIAL STATEMENT STRUCTURE

Our simplified P&L will be:

```

Revenue
(-) Cost of Goods Sold
-----
Gross Profit
Gross Margin %

(-) Operating Expenses
-----
Operating Income / EBIT
EBIT Margin %

```

This is the central financial structure.

12. UNDERSTAND THESE TERMS BEFORE BUILDING

You do not need to memorize definitions.

You need to understand the relationships.

Revenue

Money generated from selling products/services.

For devices:

$$\text{Units} \times \text{Price}$$

For subscriptions:

$$\text{Subscribers} \times \text{Price}$$

COGS

Costs directly associated with delivering products/services.

Gross Profit

$$\text{Revenue} - \text{COGS}$$

Gross Margin

$$\text{Gross Profit} / \text{Revenue}$$

Operating Expenses

Costs required to operate the company that are not included in COGS.

Examples:

- Sales
 - Marketing
 - R&D
 - IT
 - HR
 - Facilities
-

EBIT

Gross Profit - Operating Expenses

Budget

What management planned.

Actual

What the business recorded.

In this project, "actual" means **simulated actual performance**, not real company data.

Forecast

What management now expects to happen.

Variance

Difference between two financial values.

Basic variance:

Actual - Budget

13. FAVORABLE VS UNFAVORABLE VARIANCE

This is critical.

For revenue:

Actual > Budget
= Favorable

For expenses:

Actual > Budget
= Unfavorable

Therefore, never simply say:

"Positive variance is good."

It depends on what you're measuring.

14. PROJECT TIMEFRAME

Use:

Historical period

January 2025 – December 2025

Simulated actuals.

Planning year

January 2026 – December 2026

Annual budget.

Current-year actuals

January 2026 – June 2026

Simulated actuals.

Forecast period

July 2026 – December 2026

Forecast.

This gives us:

```
2025
12 months actual
    ↓
2026 Budget
12 months
    ↓
2026 Actual
Jan-Jun
    ↓
2026 Forecast
Jul-Dec
```

This is enough to demonstrate a realistic FP&A cycle.

15. PROJECT STATUS DATE

For the purposes of the case study:

Planning / reporting date: June 30, 2026

This means June is the latest closed month.

July–December are forecast.

This makes the project internally consistent.

16. TOOL SETUP

You currently have:

- Excel for the web
- Windows
- Microsoft account
- Excel skill level approximately 2
- SQL skill level 0
- Power BI skill level 0

That is fine.

Excel

Continue using Excel for the web.

Excel for the web supports formulas such as `SUMIFS`, PivotTables, charts, and slicers.

However, this project deliberately avoids making desktop-only Excel features a dependency.

We will **not require**:

- VBA
- macros
- Power Pivot
- complicated external connections
- Excel desktop-only automation

This keeps the project compatible with your setup.

17. SQL SOFTWARE

Use:

SQL Server 2025 Express

It is free and intended for learning/development and smaller data-driven applications. SQL Server 2025 Express has a 50 GB maximum relational database size, which is vastly more than we need.

Use:

SQL Server Management Studio 22

SSMS is Microsoft's free Windows application for working with SQL Server and includes the query editor we need.

Official resources:

[SQL Server downloads](https://www.microsoft.com/en-us/sql-server/sql-server-downloads)

[SQL Server installation guide](#)

[SQL Server Management Studio](#)

18. POWER BI SOFTWARE

Use:

Power BI Desktop

Power BI Desktop is free to download and use for creating reports and data models. Publishing/sharing through the Power BI service can require paid licensing, but that is not required for this portfolio project.

Official download:

 [Power BI Desktop download](#) 

You only need Power BI Desktop.

Do not worry about Power BI Service yet.

19. PROJECT FOLDER STRUCTURE

Create this folder on your computer:

```
MadeUP-FPA-Project/  
|  
├── 01_Business_Requirements/  
|  
├── 02_Data/  
|   ├── Raw/  
|   ├── SQL_Exports/  
|   └── PowerBI_Exports/  
|  
├── 03_SQL/  
|  
├── 04_Excel/  
|  
├── 05_PowerBI/  
|  
├── 06_Documentation/  
|  
├── 07_Screenshots/  
|  
└── 08_Project_Log/
```

This structure should remain throughout the project.

Do not randomly save files on your desktop.

20. CREATE THE PROJECT LOG

Create:

```
08_Project_Log/PROJECT_PROGRESS.md
```

Use this structure:

```
# MadeUP FP&A Project Progress

## Current Status
Phase:

## Completed
-

## What I Learned
-

## What I Built
-

## Skills Demonstrated
-

## Problems Encountered
-

## How I Solved Them
-

## Interview Notes
-

## Next Step
-
```

Update this after every major phase.

This is important.

You are not only building a portfolio.

You are building your **interview memory**.

21. PHASE 0 — CREATE THE GITHUB REPOSITORY

Repository name:

MadeUP-FPA-Project

Description:

An end-to-end FP&A analytics case study for MadeUP, a fictional global technology company, using SQL, Microsoft Excel, and Power BI for financial data analysis, budgeting, forecasting, variance analysis, and management reporting.

Set:

Public

Add:

README.md

GitHub's official documentation supports creating a public repository and adding a README directly through the browser.

Do not worry about Git commands yet.

You can upload files through the GitHub website. GitHub currently allows browser uploads up to 25 MiB per file; larger files can be handled through Git/Git LFS if necessary.

22. CHECKPOINT — GITHUB

You are done when:

- ☐ Repository exists
- ☐ Repository is public

- [] Repository is named `MadeUP-FPA-Project`
- [] README exists
- [] Project folders are planned

You learned

Basic repository structure and portfolio organization.

Portfolio evidence

You now have a public home for the project.

23. PHASE 1 — BUSINESS REQUIREMENTS

Before SQL.

Before Excel.

Before Power BI.

Document the business.

Create:

```
01_Business_Requirements/Business_Model.md
```

Include:

Company

MadeUP

Industry

Technology

Business Units

- Devices
- Cloud & Software
- Digital Services

Regions

- North America
- Europe
- Asia Pacific
- Latin America

Departments

- Sales
- Marketing
- R&D
- G&A
- IT
- Customer Operations
- HR
- Facilities

Financial objectives

Management wants to know:

1. Are we meeting budget?
2. Which products are driving revenue?
3. Which regions are underperforming?
4. Where are expenses exceeding budget?
5. What is the latest full-year forecast?
6. What are the largest financial risks?
7. What are the largest opportunities?
8. What actions should management consider?

24. PHASE 1.1 — DEFINE KPIS

Create:

01_Business_Requirements/KPI_Definitions.md

Use:

KPI	Formula	Purpose
Revenue	Product/Service Revenue	Top-line performance
Revenue Growth %	Current / Prior - 1	Growth

KPI	Formula	Purpose
COGS	Cost of goods sold	Direct cost
Gross Profit	Revenue - COGS	Profit after direct costs
Gross Margin %	Gross Profit / Revenue	Margin quality
Operating Expense	Sum of Opex	Operating cost
EBIT	Gross Profit - Opex	Operating profitability
EBIT Margin %	EBIT / Revenue	Profitability
Budget Variance	Actual - Budget	Performance
Variance %	Variance / Budget	Relative performance
Forecast	Actual YTD + Forecast Remaining	Latest outlook
Headcount	Active employees	Workforce scale

25. CHECKPOINT — BUSINESS MODEL

Before continuing, you should be able to explain:

"MadeUP is a technology company with three business units. Devices generate revenue through unit sales, cloud/software generates subscription revenue, and digital services generate subscription/transaction revenue. The company operates across four regions and has eight major functional departments."

If you cannot explain that naturally, stop here and review.

You learned

How an FP&A analyst starts with the business rather than the database.

FP&A capability

Business understanding.

26. PHASE 2 — SQL DATABASE

Now we build the data layer.

The database will contain:

DimDate
DimProduct
DimRegion
DimDepartment
DimExpenseCategory

FactRevenueActual
FactHeadcountActual
FactOpexActual

Think:

Dimension

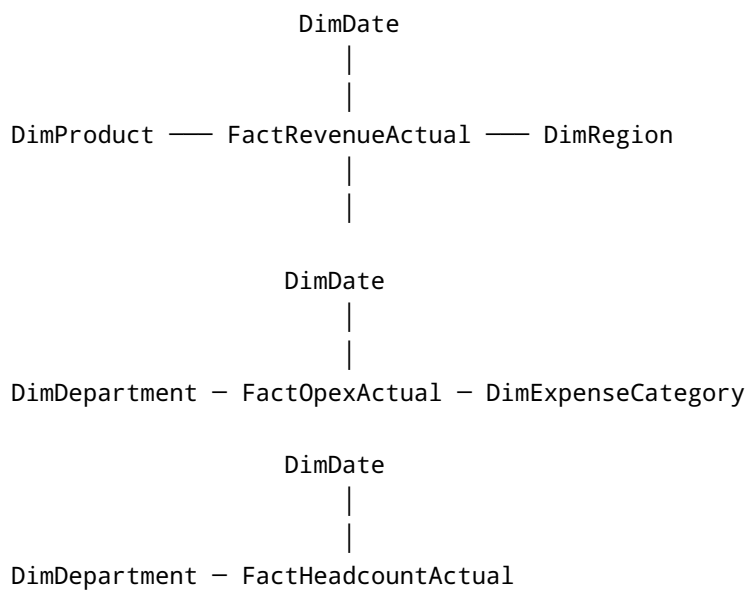
Describes something.

Fact

Contains measurable business activity.

27. THE STAR SCHEMA

Your conceptual model is:



This is a simplified star-schema approach.

28. CREATE THE DATABASE

Open SSMS.

Connect to your local SQL Server instance.

Create a new query.

Run:

```
CREATE DATABASE MadeUP_FPA;  
GO
```

Then:

```
USE MadeUP_FPA;  
GO
```

If successful, you now have your project database.

29. WHAT YOU JUST LEARNED

A database is a structured storage system.

SQL is not Excel with different syntax.

Excel is primarily a spreadsheet environment.

SQL works with structured relational data.

The database is our:

Financial data layer.

30. CREATE DIMENSION TABLES

Start with:

```
CREATE TABLE DimProduct (  
    ProductID VARCHAR(10) PRIMARY KEY,  
    ProductName VARCHAR(100) NOT NULL,  
    BusinessUnit VARCHAR(50) NOT NULL,  
    RevenueModel VARCHAR(30) NOT NULL,  
    BasePrice DECIMAL(12,2) NOT NULL,  
    COGSRate DECIMAL(5,4) NOT NULL  
);  
GO
```

Then:

```
CREATE TABLE DimRegion (  
    RegionID VARCHAR(10) PRIMARY KEY,  
    RegionName VARCHAR(50) NOT NULL,  
    RegionFactor DECIMAL(8,4) NOT NULL  
);  
GO
```

Then:

```
CREATE TABLE DimDepartment (  
    DepartmentID VARCHAR(10) PRIMARY KEY,  
    DepartmentName VARCHAR(100) NOT NULL,  
    BaseHeadcount INT NOT NULL,  
    BaseMonthlySalary DECIMAL(12,2) NOT NULL,  
    BaseNonPayrollOpex DECIMAL(14,2) NOT NULL  
);  
GO
```

Then:

```
CREATE TABLE DimExpenseCategory (  
    ExpenseCategoryID VARCHAR(10) PRIMARY KEY,  
    ExpenseCategoryName VARCHAR(100) NOT NULL  
);  
GO
```

31. INSERT PRODUCT DATA

Run:

```
INSERT INTO DimProduct
(ProductID, ProductName, BusinessUnit, RevenueModel, BasePrice, COGSRate)
VALUES
('P001', 'MadeUP Phone', 'Devices', 'Unit Sales', 699, 0.62),
('P002', 'MadeUP Phone Pro', 'Devices', 'Unit Sales', 999, 0.58),
('P003', 'MadeUP Pad', 'Devices', 'Unit Sales', 549, 0.60),
('P004', 'MadeUP Book', 'Devices', 'Unit Sales', 1199, 0.55),
('P005', 'MadeUP Watch', 'Devices', 'Unit Sales', 299, 0.50),
('P006', 'MadeUP Cloud', 'Cloud & Software', 'Subscription', 39, 0.25),
('P007', 'MadeUP Workspace', 'Cloud & Software', 'Subscription', 24, 0.20),
('P008', 'MadeUP Music', 'Digital Services', 'Subscription', 12, 0.30),
('P009', 'MadeUP Video', 'Digital Services', 'Subscription', 15, 0.35),
('P010', 'MadeUP Store', 'Digital Services', 'Transaction', 8, 0.15);
GO
```

Check:

```
SELECT *
FROM DimProduct;
```

32. INSERT REGIONS

```
INSERT INTO DimRegion
(RegionID, RegionName, RegionFactor)
VALUES
('R01', 'North America', 1.00),
('R02', 'Europe', 0.85),
('R03', 'Asia Pacific', 0.95),
('R04', 'Latin America', 0.55);
GO
```

Check:

```
SELECT *
FROM DimRegion;
```

33. INSERT DEPARTMENTS

Use:

```
INSERT INTO DimDepartment
(DepartmentID, DepartmentName, BaseHeadcount, BaseMonthlySalary,
BaseNonPayrollOpex)
VALUES
('D01', 'Sales', 700, 70000, 8000000),
('D02', 'Marketing', 450, 75000, 30000000),
('D03', 'Research & Development', 1100, 95000, 20000000),
('D04', 'General & Administrative', 400, 85000, 8000000),
('D05', 'Information Technology', 350, 90000, 12000000),
('D06', 'Customer Operations', 700, 65000, 10000000),
('D07', 'Human Resources', 150, 70000, 3000000),
('D08', 'Facilities', 200, 55000, 8000000);
GO
```

34. INSERT EXPENSE CATEGORIES

```
INSERT INTO DimExpenseCategory
(ExpenseCategoryID, ExpenseCategoryName)
VALUES
('A01', 'Payroll'),
('A02', 'Operating Programs'),
('A03', 'Technology & Software'),
('A04', 'Facilities & Travel');
GO
```

35. CREATE DATE TABLE

Create:

```
CREATE TABLE DimDate (
    MonthStart DATE PRIMARY KEY,
    CalendarYear INT NOT NULL,
    MonthNumber INT NOT NULL,
```

```
MonthName VARCHAR(20) NOT NULL,  
QuarterName VARCHAR(5) NOT NULL  
);  
GO
```

Populate 2025–2026:

```
WITH Months AS (  
    SELECT CAST('2025-01-01' AS DATE) AS MonthStart  
  
    UNION ALL  
  
    SELECT DATEADD(MONTH,1,MonthStart)  
    FROM Months  
    WHERE MonthStart < '2026-06-01'  
)  
INSERT INTO DimDate  
(MonthStart, CalendarYear, MonthNumber, MonthName, QuarterName)  
SELECT  
    MonthStart,  
    YEAR(MonthStart),  
    MONTH(MonthStart),  
    DATENAME(MONTH,MonthStart),  
    CONCAT('Q',DATEPART(QUARTER,MonthStart))  
FROM Months  
OPTION (MAXRECURSION 100);  
GO
```

Check:

```
SELECT *  
FROM DimDate  
ORDER BY MonthStart;
```

You should have:

18 rows

January 2025 through June 2026.

36. CREATE REVENUE FACT TABLE

```
CREATE TABLE FactRevenueActual (  
    MonthStart DATE NOT NULL,  
    ProductID VARCHAR(10) NOT NULL,  
    RegionID VARCHAR(10) NOT NULL,  
    Units DECIMAL(18,2) NOT NULL,  
    ASP DECIMAL(12,2) NOT NULL,  
    Revenue DECIMAL(18,2) NOT NULL,  
    COGS DECIMAL(18,2) NOT NULL,  
  
    CONSTRAINT PK_FactRevenueActual  
        PRIMARY KEY (MonthStart, ProductID, RegionID),  
  
    CONSTRAINT FK_Revenue_Product  
        FOREIGN KEY (ProductID)  
        REFERENCES DimProduct(ProductID),  
  
    CONSTRAINT FK_Revenue_Region  
        FOREIGN KEY (RegionID)  
        REFERENCES DimRegion(RegionID),  
  
    CONSTRAINT FK_Revenue_Date  
        FOREIGN KEY (MonthStart)  
        REFERENCES DimDate(MonthStart)  
);  
GO
```

37. REVENUE GENERATION LOGIC

We need synthetic but realistic behavior.

Base monthly volumes

Use approximately:

Product	Base Monthly Units
Phone	250,000
Phone Pro	120,000
Pad	140,000

Product	Base Monthly Units
Book	70,000
Watch	180,000
Cloud	600,000
Workspace	400,000
Music	1,200,000
Video	800,000
Store	900,000

The numbers represent units, subscribers, or transactions depending on the product.

38. SEASONALITY

Technology companies don't necessarily sell the same amount every month.

Use:

Month	Factor
Jan	0.92
Feb	0.90
Mar	0.96
Apr	0.95
May	1.00
Jun	1.02
Jul	1.03
Aug	1.02
Sep	1.00
Oct	1.05
Nov	1.12
Dec	1.18

This creates a stronger Q4.

39. REVENUE ACTUAL GENERATION

Use the following conceptual formula:

```
Units
=
Base Units
× Region Factor
× Seasonality
× Growth Factor
× Actual Performance Factor
```

Then:

```
Revenue
=
Units × ASP
```

and:

```
COGS
=
Revenue × COGS Rate
```

The SQL generation script can implement those calculations.

Do not spend excessive time manually inserting thousands of rows.

The purpose is to **learn SQL and build the analytical environment**, not manually type data.

40. CREATE HEADCOUNT FACT

```
CREATE TABLE FactHeadcountActual (
    MonthStart DATE NOT NULL,
    DepartmentID VARCHAR(10) NOT NULL,
    Headcount DECIMAL(12,2) NOT NULL,
    AverageMonthlySalary DECIMAL(12,2) NOT NULL,

    CONSTRAINT PK_FactHeadcountActual
    PRIMARY KEY (MonthStart, DepartmentID),
```

```

CONSTRAINT FK_Headcount_Date
    FOREIGN KEY (MonthStart)
    REFERENCES DimDate(MonthStart),

CONSTRAINT FK_Headcount_Department
    FOREIGN KEY (DepartmentID)
    REFERENCES DimDepartment(DepartmentID)

);
GO

```

41. CREATE OPEX FACT

```

CREATE TABLE FactOpexActual (
    MonthStart DATE NOT NULL,
    DepartmentID VARCHAR(10) NOT NULL,
    ExpenseCategoryID VARCHAR(10) NOT NULL,
    Amount DECIMAL(18,2) NOT NULL,

    CONSTRAINT PK_FactOpexActual
        PRIMARY KEY (MonthStart, DepartmentID, ExpenseCategoryID),

    CONSTRAINT FK_Opex_Date
        FOREIGN KEY (MonthStart)
        REFERENCES DimDate(MonthStart),

    CONSTRAINT FK_Opex_Department
        FOREIGN KEY (DepartmentID)
        REFERENCES DimDepartment(DepartmentID),

    CONSTRAINT FK_Opex_Category
        FOREIGN KEY (ExpenseCategoryID)
        REFERENCES DimExpenseCategory(ExpenseCategoryID)

);
GO

```

42. DATA VALIDATION — SQL

Before touching Excel, run:

```
SELECT COUNT(*) AS ProductCount
FROM DimProduct;
```

Expected:

10

Then:

```
SELECT COUNT(*) AS RegionCount
FROM DimRegion;
```

Expected:

4

Then:

```
SELECT COUNT(*) AS DepartmentCount
FROM DimDepartment;
```

Expected:

8

Then:

```
SELECT MIN(MonthStart) AS FirstMonth,
       MAX(MonthStart) AS LastMonth,
       COUNT(*) AS MonthCount
FROM DimDate;
```

Expected:

2025-01-01
2026-06-01
18

43. FIRST SQL ANALYSIS

Run:

```
SELECT
    YEAR(MonthStart) AS CalendarYear,
    SUM(Revenue) AS Revenue
FROM FactRevenueActual
GROUP BY YEAR(MonthStart)
ORDER BY CalendarYear;
```

You just performed a financial analysis query.

44. MONTHLY REVENUE

```
SELECT
    MonthStart,
    SUM(Revenue) AS Revenue
FROM FactRevenueActual
GROUP BY MonthStart
ORDER BY MonthStart;
```

This answers:

How much revenue did MadeUP generate each month?

45. REVENUE BY BUSINESS UNIT

```
SELECT
    p.BusinessUnit,
    SUM(f.Revenue) AS Revenue
FROM FactRevenueActual f
JOIN DimProduct p
    ON f.ProductID = p.ProductID
GROUP BY p.BusinessUnit
ORDER BY Revenue DESC;
```

This is your first important `JOIN`.

You are combining:

Fact data

with:

Product attributes.

46. REVENUE BY REGION

```
SELECT
    r.RegionName,
    SUM(f.Revenue) AS Revenue
FROM FactRevenueActual f
JOIN DimRegion r
    ON f.RegionID = r.RegionID
GROUP BY r.RegionName
ORDER BY Revenue DESC;
```

47. REVENUE BY PRODUCT

```
SELECT
    p.ProductName,
    SUM(f.Revenue) AS Revenue
FROM FactRevenueActual f
JOIN DimProduct p
    ON f.ProductID = p.ProductID
GROUP BY p.ProductName
ORDER BY Revenue DESC;
```

48. COGS AND GROSS PROFIT

```
SELECT
    SUM(Revenue) AS Revenue,
    SUM(COGS) AS COGS,
    SUM(Revenue - COGS) AS GrossProfit,
```

```
SUM(Revenue - COGS) / NULLIF(SUM(Revenue),0) AS GrossMargin
FROM FactRevenueActual;
```

You have now performed a basic P&L calculation in SQL.

49. OPEX ANALYSIS

```
SELECT
    d.DepartmentName,
    SUM(o.Amount) AS Opex
FROM FactOpexActual o
JOIN DimDepartment d
    ON o.DepartmentID = d.DepartmentID
GROUP BY d.DepartmentName
ORDER BY Opex DESC;
```

50. TOP OPEX CATEGORIES

```
SELECT
    e.ExpenseCategoryName,
    SUM(o.Amount) AS Opex
FROM FactOpexActual o
JOIN DimExpenseCategory e
    ON o.ExpenseCategoryID = e.ExpenseCategoryID
GROUP BY e.ExpenseCategoryName
ORDER BY Opex DESC;
```

51. SQL CHECKPOINT

You should now understand:

- [] What a database is
- [] What a table is
- [] What a primary key is
- [] What a foreign key is
- [] What a dimension table is
- [] What a fact table is
- [] What a JOIN does

- [] What GROUP BY does
- [] What SUM does
- [] How SQL can answer financial questions

Skill achieved

SQL fundamentals + financial data analysis

Resume evidence

You can legitimately demonstrate SQL queries against a structured financial database.

52. EXPORT SQL RESULTS

We need to move SQL output into the Excel stage.

SSMS can save query results to files, including CSV.

For example, run:

```
SELECT *  
FROM FactRevenueActual  
ORDER BY MonthStart, ProductID, RegionID;
```

Save the results as:

```
02_Data/SQL_Exports/Actual_Revenue.csv
```

Repeat for:

```
Actual_Opex.csv  
Actual_Headcount.csv
```

Also export the dimension tables:

```
Products.csv  
Regions.csv  
Departments.csv  
ExpenseCategories.csv
```

53. SQL EXPORT CHECKPOINT

Your folder should now contain:

```
02_Data/  
└─ SQL_Exports/  
    └─ Actual_Revenue.csv  
    └─ Actual_Opex.csv  
    └─ Actual_Headcount.csv  
    └─ Products.csv  
    └─ Regions.csv  
    └─ Departments.csv  
    └─ ExpenseCategories.csv
```

Important

Do not manually edit these CSVs after exporting them.

SQL is the source.

54. PHASE 3 — EXCEL FP&A MODEL

Now Excel becomes the main FP&A environment.

Create:

```
04_Excel/MadeUP_FPA_Model.xlsx
```

Since you are using Excel for the web, save it in OneDrive.

55. EXCEL WORKBOOK STRUCTURE

Create these sheets:

```
00_ReadMe  
01_Assumptions  
02_Actual_Revenue  
03_Actual_Opex
```

04_Actual_Headcount
05_Budget
06_Forecast
07_PnL
08_Variance
09_Scenarios
10_Management_Summary
11_Checks

56. 00_README

Include:

Project

MadeUP FP&A Model

Reporting Date

June 30, 2026

Historical Actuals

January 2025 – December 2025

Current-Year Actuals

January 2026 – June 2026

Forecast

July 2026 – December 2026

Data

Synthetic

Purpose

Demonstrate an end-to-end FP&A workflow.

57. 01_ASSUMPTIONS

This is one of the most important sheets.

Create sections:

Revenue assumptions

Business Unit	Volume Growth	Price Growth
Devices	6%	2%
Cloud & Software	10%	3%
Digital Services	8%	2%

Opex assumptions

Department	Budget Growth
Sales	6%
Marketing	7%
R&D	9%
G&A	5%
IT	8%
Customer Operations	6%
HR	5%
Facilities	5%

Headcount assumptions

Department	Headcount Growth
Sales	5%
Marketing	6%
R&D	8%
G&A	3%

Department	Headcount Growth
IT	7%
Customer Operations	5%
HR	4%
Facilities	2%

58. WHY ASSUMPTIONS MATTER

This is one of the most important FP&A concepts.

A weak model says:

$$\text{Revenue in 2026} = \text{Revenue in 2025} \times 1.06$$

A stronger model asks:

What drives revenue?

For devices:

$$\text{Volume} \times \text{Price}$$

For subscriptions:

$$\text{Subscribers} \times \text{Price}$$

This is called:

Driver-based planning.

That is what we want to demonstrate.

59. IMPORT ACTUAL REVENUE

Open `02_Actual_Revenue`.

Import/paste the SQL CSV into the worksheet.

Turn it into an Excel Table.

Use a clear table name:

tblActualRevenue

Columns:

MonthStart
ProductID
RegionID
Units
ASP
Revenue
COGS

Do not modify the actual numbers.

60. IMPORT ACTUAL OPEX

Use:

tblActualOpex

Columns:

MonthStart
DepartmentID
ExpenseCategoryID
Amount

61. IMPORT ACTUAL HEADCOUNT

Use:

tblActualHeadcount

Columns:

```
MonthStart
DepartmentID
Headcount
AverageMonthlySalary
```

62. BUDGET MODEL — REVENUE

The budget will use 2025 actuals as the base.

For each product-region-month:

```
Budget Units
=
2025 Actual Units
×
(1 + Volume Growth)
```

Then:

```
Budget ASP
=
2025 Actual ASP
×
(1 + Price Growth)
```

Then:

```
Budget Revenue
=
Budget Units × Budget ASP
```

Then:

```
Budget COGS
=
Budget Revenue × COGS Rate
```

63. XLOOKUP

You will use `XLOOKUP` to retrieve business information.

Example:

```
=XLOOKUP([@ProductID],Products[ProductID],Products[BusinessUnit])
```

This teaches you how to connect a transaction/model row with its master data.

Microsoft recommends XLOOKUP as a modern lookup function, and it is available in current Microsoft 365 Excel environments.

64. SUMIFS

You will use `SUMIFS` heavily.

For example:

```
=SUMIFS(  
    tblActualRevenue[Revenue],  
    tblActualRevenue[MonthStart],A2,  
    tblActualRevenue[ProductID],B2,  
    tblActualRevenue[RegionID],C2  
)
```

This means:

Find the revenue matching this month, product, and region.

That is a core FP&A modeling technique.

65. BUILD THE 2026 BUDGET TABLE

Create:

05_Budget

Columns:

Month
ProductID
RegionID
BusinessUnit
Budget Units
Budget ASP
Budget Revenue
Budget COGS
Budget Gross Profit
Budget Gross Margin

Use January–December 2026.

66. BUDGET VALIDATION

At the bottom:

Total Budget Revenue
Total Budget COGS
Total Budget Gross Profit
Budget Gross Margin %

Then check:

Revenue - COGS = Gross Profit

and:

Gross Profit / Revenue = Gross Margin

67. BUDGET OPEX

Create an Opex section.

Formula:

```
2026 Budget Opex
=
2025 Actual Opex
×
(1 + Department Growth Assumption)
```

Use `SUMIFS` to retrieve the corresponding 2025 amount.

68. BUDGET HEADCOUNT

Formula:

```
Budget Headcount
=
2025 Actual Headcount
×
(1 + Headcount Growth)
```

Round appropriately.

For example:

```
=ROUND(BaseHeadcount*(1+GrowthRate),0)
```

69. BUILD THE BUDGET P&L

Your P&L should eventually look like:

\$M	2026 Budget
Revenue	
COGS	
Gross Profit	
Gross Margin	

\$M	2026 Budget
Operating Expenses	
EBIT	
EBIT Margin	

Then break the model down by month.

70. PHASE 4 — ACTUAL VS BUDGET

Create:

08_Variance

For each month:

Actual
Budget
Variance \$
Variance %
Status

Revenue:

=Actual-Budget

Variance %:

=IFERROR((Actual-Budget)/Budget,0)

Status:

=IF(Variance>=0,"Favorable","Unfavorable")

For expenses:

```
=IF(Actual<=Budget,"Favorable","Unfavorable")
```

71. MATERIALITY

FP&A analysts don't investigate every \$100 difference.

Create a materiality rule.

For this project:

Material = variance greater than 5% OR absolute variance greater than \$5M.

This is a project assumption.

It is not a universal corporate policy.

Use:

```
=IF(  
  OR(ABS(Variance%)>=5%,ABS(Variance$)>=5000000),  
  "Material",  
  "Not Material"  
)
```

This gives you a professional control.

72. BUILD VARIANCE ANALYSIS

Analyze:

Revenue

- By business unit
- By product
- By region
- By month

Opex

- By department

- By expense category
 - By month
-

73. EXAMPLE MANAGEMENT QUESTION

Your model should answer:

Why is revenue below budget?

Break it down:

```
Total Revenue Variance
      ↓
Business Unit
      ↓
Region
      ↓
Product
```

This is called:

Variance driver analysis.

74. BUILD THE FORECAST

Create:

```
06_Forecast
```

Forecast period:

July–December 2026

Use a simple and defensible methodology.

75. FORECAST METHOD

We will use:

YTD run-rate forecast

This is intentionally simple.

For each major financial line:

$$\begin{array}{l} \text{YTD Actual} \\ \div \\ \text{YTD Budget} \\ = \\ \text{YTD Performance Ratio} \end{array}$$

Example:

Actual = \$510M

Budget = \$525M

$$510 / 525 = 97.1\%$$

Then:

$$\begin{array}{l} \text{Remaining Budget} \\ \times \\ 97.1\% \\ = \\ \text{Forecast Remaining} \end{array}$$

Finally:

$$\begin{array}{l} \text{Full-Year Forecast} \\ = \\ \text{YTD Actual} \\ + \\ \text{Forecast Remaining} \end{array}$$

76. WHY THIS METHOD

Because the objective is not to demonstrate advanced econometrics.

The objective is to demonstrate:

- understanding of forecasting
- use of actual results
- use of budget assumptions
- forecast updating
- financial judgment

Later, you can improve the forecast methodology.

For the core portfolio project, this is enough.

77. FORECAST P&L

Create:

\$M	Budget	YTD Actual	Forecast Remaining	FY Forecast	Variance
Revenue					
COGS					
Gross Profit					
Gross Margin					
Opex					
EBIT					
EBIT Margin					

78. BUDGET VS FORECAST

This is an important FP&A output.

Management wants:

"What do we think the full year will look like now?"

Show:

FY Budget
vs
FY Forecast

Then calculate:

Forecast Variance
=
Forecast - Budget

79. SCENARIO ANALYSIS

Create:

09_Scenarios

Build three scenarios:

Downside

Revenue growth assumptions reduced.

Base

Current assumptions.

Upside

Revenue growth assumptions increased.

Use:

Scenario	Revenue Growth Adjustment
Downside	-3 percentage points
Base	0
Upside	+3 percentage points

Then calculate:

- Revenue
 - Gross Profit
 - Gross Margin
 - EBIT
 - EBIT Margin
-

80. SCENARIO OUTPUT

Your model should show:

\$M	Downside	Base	Upside
Revenue			
Gross Profit			
Gross Margin			
EBIT			
EBIT Margin			

81. WHAT MANAGEMENT SHOULD LEARN

The model should answer:

What happens if revenue growth is weaker than expected?

What happens if growth exceeds expectations?

How much does EBIT change?

This demonstrates:

Scenario analysis and decision support.

82. PHASE 5 — MANAGEMENT SUMMARY IN EXCEL

Create:

10_Management_Summary

This is not a giant dashboard.

Keep it professional.

Top KPIs:

Revenue
Revenue vs Budget
Gross Margin
EBIT
EBIT vs Budget
FY Forecast
Forecast vs Budget
Headcount

Then:

Top Risks

Example:

- Europe revenue below budget
- Device mix unfavorable
- R&D spending above plan

Top Opportunities

Example:

- APAC cloud growth
- Digital services subscriptions
- Higher-margin software growth

Recommended Actions

Example:

- Review Europe device performance
- Reassess product mix assumptions
- Protect cloud/software investment
- Monitor R&D spending against milestones

The actual findings must come from your completed model.

Do not invent them before seeing the results.

83. EXCEL MODEL CONTROLS

Create:

11_Checks

Use checks such as:

Revenue check

Detailed Revenue
=
P&L Revenue

COGS check

Detailed COGS
=
P&L COGS

Gross Profit check

Revenue - COGS
=
Gross Profit

EBIT check

Gross Profit - Opex
=
EBIT

Forecast check

YTD Actual + Forecast Remaining
=
Full-Year Forecast

Every check should return:

PASS

or:

FAIL

84. PROFESSIONAL MODEL CONTROL

Create a master check:

=IF(COUNTIF(CheckRange,"FAIL")=0,"MODEL PASS","MODEL ERROR")

Your model should not be considered complete until:

MODEL PASS

85. EXCEL CHECKPOINT

You should now have:

- ☐ Actual financial data
- ☐ Assumptions
- ☐ 2026 budget
- ☐ P&L
- ☐ Budget vs Actual
- ☐ Variance analysis
- ☐ Forecast
- ☐ Budget vs Forecast
- ☐ Scenario analysis
- ☐ Management summary

- [] Model controls

Skills achieved

- Financial modeling
- Budgeting
- Forecasting
- Variance analysis
- Scenario analysis
- Management reporting
- Advanced Excel formulas

Resume evidence

This is the primary evidence supporting:

Advanced Microsoft Excel

86. PHASE 6 — POWER BI

Now we communicate the results.

Do not start Power BI before Excel is reasonably complete.

Otherwise you will build visuals around numbers that may change.

87. POWER BI REPORT STRUCTURE

Create:

MadeUP FP&A Dashboard

Pages:

Page 1

Executive Overview

Page 2

Revenue Performance

Page 3

Budget vs Actual

Page 4

Forecast & Outlook

Page 5

Opex & Headcount

Page 6

Variance Drivers

88. DATA SOURCES

For portfolio portability, Power BI should ultimately use exported files rather than depending on your personal SQL Server installation.

Create:

02_Data/PowerBI_Exports/

Export the final reporting tables from your SQL/Excel workflow into this folder.

The Power BI report should be able to use these files.

This is a deliberate portfolio design decision:

SQL is the analytical database layer; exported datasets are the portable reporting handoff.

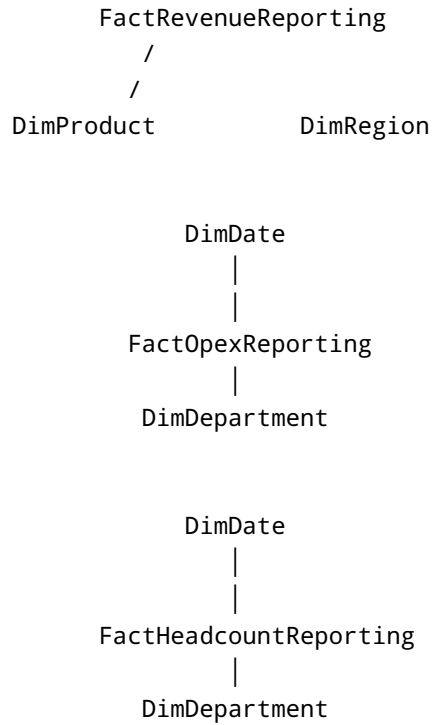
89. POWER BI DATA MODEL

Aim for:

DimDate

|

|



Do not create unnecessary relationships.

90. POWER BI — IMPORT DATA

Open Power BI Desktop.

Choose:

Get Data

Import your CSV/Excel reporting files.

For each table:

- verify column names
- verify data types
- verify dates
- verify numbers
- verify text fields

91. POWER QUERY

Use Power Query for simple data preparation:

- rename columns
- remove unnecessary columns
- change data types
- remove blank rows
- standardize field names

Do not spend hours doing complex transformations.

The project is about FP&A, not Power Query engineering.

92. DATA TYPES

Make sure:

Dates

Date

Revenue

Decimal number

Units

Whole/decimal number

Percentages

Decimal number

Product/Region/Department

Text

Incorrect data types are one of the easiest ways to break a BI model.

93. DAX MEASURES

Create measures.

Start with:

```
Revenue =  
SUM(FactRevenueReporting[Revenue])
```

Then:

```
COGS =  
SUM(FactRevenueReporting[COGS])
```

Then:

```
Gross Profit =  
[Revenue] - [COGS]
```

Then:

```
Gross Margin % =  
DIVIDE([Gross Profit], [Revenue])
```

94. OPEX MEASURE

```
Operating Expenses =  
SUM(FactOpexReporting[Amount])
```

Then:

```
EBIT =  
[Gross Profit] - [Operating Expenses]
```

Then:

```
EBIT Margin % =  
DIVIDE([EBIT], [Revenue])
```

95. BUDGET MEASURES

```
Budget Revenue =  
SUM(FactBudgetRevenue[BudgetRevenue])
```

Then:

```
Revenue Variance =  
[Revenue] - [Budget Revenue]
```

Then:

```
Revenue Variance % =  
DIVIDE(  
    [Revenue Variance],  
    [Budget Revenue]  
)
```

96. FAVORABLE VARIANCE

For revenue:

```
Revenue Favorable =  
[Revenue] - [Budget Revenue]
```

For Opex:

```
Opex Favorable =  
[Budget Opex] - [Operating Expenses]
```

This keeps financial interpretation correct.

97. EXECUTIVE OVERVIEW PAGE

Your first Power BI page should show:

KPI cards

- Revenue
- Revenue Variance %
- Gross Margin %
- EBIT
- EBIT Margin %
- FY Forecast
- Forecast Variance %

Then:

Charts

1. Revenue Actual vs Budget by Month
2. Revenue by Business Unit
3. Revenue by Region
4. EBIT trend

Then:

Management callouts

- Biggest unfavorable variance
 - Biggest favorable variance
 - Biggest risk
 - Biggest opportunity
-

98. REVENUE PERFORMANCE PAGE

Show:

Revenue by:

- Product
- Business Unit
- Region
- Month

Recommended visuals:

- line chart
- clustered bar chart
- matrix
- KPI cards

Avoid visual overload.

99. BUDGET VS ACTUAL PAGE

Show:

Actual
Budget
Variance \$
Variance %

By:

- month
- business unit
- region
- product

This should allow management to identify where performance differs from plan.

100. FORECAST PAGE

Show:

Full-year budget

vs.

Full-year forecast

Then:

- Revenue outlook
- EBIT outlook
- Margin outlook

- Forecast by business unit
 - Forecast variance
-

101. OPEX PAGE

Show:

Opex by department

Opex by category

Opex vs Budget

Headcount

Payroll

Useful questions:

Which department has the largest cost?

Which department is over budget?

Is headcount growth contributing to expense growth?

102. VARIANCE DRIVER PAGE

This is one of the most valuable pages.

Show:

Top unfavorable revenue variances

Top favorable revenue variances

Top unfavorable expense variances

Top favorable expense variances

Then provide:

Variance
↓
Business Unit
↓
Region
↓
Product / Department

103. POWER BI DESIGN RULES

Do:

- use clear titles
- use consistent units
- use \$M rather than giant dollar amounts
- use percentages appropriately
- keep visual hierarchy
- use whitespace
- use meaningful labels
- keep the number of visuals controlled

Do not:

- fill every inch of the page
 - use 15 charts on one page
 - create decorative visuals with no business purpose
 - use pie charts everywhere
 - use unexplained KPIs
 - create dashboards that cannot answer a management question
-

104. POWER BI CHECKPOINT

You should be able to explain:

What is a table?

What is a relationship?

What is a measure?

What is DAX?

Why do we need a date table?

Why should revenue and budget be modeled carefully?

How does Power BI help management?

Skills achieved

- Power Query
- Data modeling
- DAX
- KPI reporting
- Financial visualization
- Management dashboards

Resume evidence

You can legitimately demonstrate hands-on Power BI work.

105. PHASE 7 — RECONCILIATION

This is mandatory.

Do not skip it.

A professional-looking dashboard with inconsistent numbers is worse than no dashboard.

106. RECONCILE SQL TO EXCEL

Compare:

SQL Revenue
vs
Excel Actual Revenue

They must match.

Then:

SQL COGS
vs
Excel Actual COGS

They must match.

Then:

SQL Opex
vs
Excel Actual Opex

They must match.

107. RECONCILE EXCEL TO POWER BI

Compare:

Excel Revenue
vs
Power BI Revenue

Then:

Excel EBIT
vs
Power BI EBIT

Then:

Excel Opex
vs
Power BI Opex

The numbers should reconcile.

108. RECONCILIATION TABLE

Create:

06_Documentation/Reconciliation.md

Use:

Metric	SQL	Excel	Power BI	Difference	Status
Revenue					PASS
COGS					PASS
Gross Profit					PASS
Opex					PASS
EBIT					PASS

109. ZERO-TOLERANCE RULE

For core financial totals:

Difference must equal zero, subject to clearly documented rounding.

Do not accept:

"It's close enough."

Financial reporting requires control.

110. PHASE 8 — MANAGEMENT ANALYSIS

Now stop looking at tools.

Pretend you are the FP&A analyst presenting to management.

Answer:

1. Are we meeting budget?
 2. What is the revenue variance?
 3. What is driving it?
 4. Which region is strongest?
 5. Which region is weakest?
 6. Which products are underperforming?
 7. Where are expenses above budget?
 8. What does the FY forecast look like?
 9. What are the largest risks?
 10. What are the largest opportunities?
 11. What should management do?
-

111. CREATE MANAGEMENT COMMENTARY

Create:

```
06_Documentation/Management_Commentary.md
```

Use this structure:

```
# Management Commentary

## Executive Summary

[3-5 sentences]

## Revenue Performance

[What happened?]

## Gross Margin

[What happened?]
```

Operating Expenses

[What happened?]

Forecast

[What is expected?]

Key Risks

- 1.
- 2.
- 3.

Key Opportunities

- 1.
- 2.
- 3.

Recommended Actions

- 1.
- 2.
- 3.

Only write conclusions after analyzing the finished model.

112. THE QUALITY OF YOUR ANALYSIS

Weak:

Revenue was below budget.

Better:

Revenue was below budget, primarily driven by weaker performance in Europe.

Strong:

Revenue was below budget, primarily driven by weaker device performance in Europe, partially offset by stronger cloud and digital-services growth in Asia Pacific. The updated full-year forecast indicates that the current shortfall is likely to persist unless second-half device volumes recover.

The third version is what you should aim for.

113. PHASE 9 — PROJECT DOCUMENTATION

Create:

```
06_Documentation/  
├─ Business_Model.md  
├─ Data_Dictionary.md  
├─ Methodology.md  
├─ KPI_Definitions.md  
├─ Reconciliation.md  
├─ Management_Commentary.md  
└─ Limitations.md
```

114. DATA DICTIONARY

Document:

Field	Description	Source	Type
MonthStart	Reporting month	SQL	Date
ProductID	Product identifier	SQL	Text
RegionID	Region identifier	SQL	Text
Revenue	Product revenue	SQL	Decimal
COGS	Direct product cost	SQL	Decimal
DepartmentID	Department identifier	SQL	Text
Amount	Opex amount	SQL	Decimal

Do this for important fields.

115. METHODOLOGY DOCUMENT

Explain:

Data

Synthetic data.

Historical period

2025.

Planning period

2026.

Current-year actual period

January–June 2026.

Forecast period

July–December 2026.

Budget methodology

2025 actuals + driver-based growth assumptions.

Forecast methodology

YTD run-rate methodology.

Scenario methodology

Revenue growth sensitivity.

116. LIMITATIONS DOCUMENT

This makes the project more professional.

State clearly:

This project uses synthetic data and is intended for educational and portfolio purposes.

The financial results do not represent a real company.

Budget and forecast assumptions are analyst-developed.

The project does not represent confidential corporate financial information.

Forecasting methodology is intentionally simplified to demonstrate core FP&A concepts rather than replicate a production forecasting system.

This prevents misleading claims.

117. PHASE 10 — GITHUB README

Your README should be the first thing a recruiter sees.

Structure it as:

```
# MadeUP – Integrated FP&A Analytics Project

## Overview

## Business Problem

## Objectives

## Tools

## Architecture

## Business Model

## FP&A Workflow

## SQL Component

## Excel Component

## Power BI Component

## Key Financial Questions

## Key Insights

## Project Structure

## Methodology
```



```
## Data Disclaimer
```

```
## Screenshots
```

```
## Skills Demonstrated
```

118. README — OPENING SECTION

Use something similar to:

```
# MadeUP – Integrated FP&A Analytics Project
```

MadeUP is a synthetic technology company created for an end-to-end Financial Planning & Analysis case study.

The project demonstrates how SQL, Microsoft Excel, and Power BI can work together to support financial data analysis, budgeting, forecasting, variance analysis, scenario planning, and management reporting.

The dataset is synthetic and does not represent the financial results of a real company.

119. README — BUSINESS PROBLEM

Explain:

Management needs a repeatable way to understand whether the company is meeting budget, identify the drivers of financial variance, update the full-year outlook, and communicate financial performance to decision-makers.

120. README — TOOL ROLES

Use:

Tool	Role
SQL	Financial data storage, querying, validation and analysis
Excel	Budgeting, forecasting, financial modeling and variance analysis
Power BI	Management reporting, KPI visualization and financial storytelling

This is one of the most important parts of the project.

121. README — ARCHITECTURE

Include:

```

Synthetic Financial Data
    ↓
SQL Database
    ↓
SQL Analysis / Exports
    ↓
Excel FP&A Model
    ↓
Budget / Actual / Forecast / Variance
    ↓
Power BI
    ↓
Management Dashboard
    ↓
Insights & Recommendations

```

122. SCREENSHOTS

Do not upload 30 screenshots.

Use approximately:

Screenshot 1

Excel financial model

Screenshot 2

Excel variance analysis

Screenshot 3

Power BI executive dashboard

Screenshot 4

Power BI variance analysis

Screenshot 5

SQL query + results

Put them in:

```
07_Screenshots/
```

Use meaningful names:

```
excel_fpna_model.png  
excel_variance_analysis.png  
powerbi_executive_dashboard.png  
powerbi_variance_drivers.png  
sql_revenue_analysis.png
```

123. GITHUB PROJECT STRUCTURE — FINAL

Your finished repository should look approximately like:

```
MadeUP-FPA-Project/  
|  
├── README.md  
|  
├── 01_Business_Requirements/  
|   ├── Business_Model.md  
|   ├── KPI_Definitions.md  
|   └── FP&A_Assumptions.md  
|
```

```
├── 02_Data/
│   ├── Raw/
│   ├── SQL_Exports/
│   └── PowerBI_Exports/
│
├── 03_SQL/
│   ├── 01_Create_Database.sql
│   ├── 02_Create_Tables.sql
│   ├── 03_Load_Data.sql
│   ├── 04_Analysis.sql
│   └── 05_Validation.sql
│
├── 04_Excel/
│   └── MadeUP_FPA_Model.xlsx
│
├── 05_PowerBI/
│   └── MadeUP_FPA_Dashboard.pbix
│
├── 06_Documentation/
│   ├── Data_Dictionary.md
│   ├── Methodology.md
│   ├── Reconciliation.md
│   ├── Management_Commentary.md
│   └── Limitations.md
│
├── 07_Screenshots/
│   ├── excel_fpna_model.png
│   ├── excel_variance_analysis.png
│   ├── powerbi_executive_dashboard.png
│   ├── powerbi_variance_drivers.png
│   └── sql_revenue_analysis.png
│
└── 08_Project_Log/
    └── PROJECT_PROGRESS.md
```

124. PHASE 11 — PORTFOLIO QUALITY REVIEW

Before calling the project finished, review every component.

Business

- [] Business model makes sense
- [] Products make sense

- ☐ Revenue models make sense
- ☐ Cost structure makes sense
- ☐ Departments make sense
- ☐ Assumptions are documented

SQL

- ☐ Database works
- ☐ Tables work
- ☐ Relationships make sense
- ☐ Queries are readable
- ☐ Queries answer financial questions
- ☐ Validation queries exist
- ☐ Exports work

Excel

- ☐ Assumptions are centralized
- ☐ Actuals are separated from assumptions
- ☐ Budget is formula-driven
- ☐ Forecast is formula-driven
- ☐ Variances are calculated
- ☐ Scenarios work
- ☐ P&L reconciles
- ☐ Model checks pass
- ☐ Formatting is professional

Power BI

- ☐ Data model works
- ☐ Measures work
- ☐ KPIs are correct
- ☐ Visuals answer questions
- ☐ Filters work
- ☐ Titles are clear
- ☐ Dashboard is not cluttered

Documentation

- ☐ README
- ☐ Data dictionary
- ☐ Methodology
- ☐ Reconciliation
- ☐ Management commentary
- ☐ Limitations
- ☐ Screenshots

125. FINAL INTERVIEW TEST

The project is NOT finished until you can answer these.

Question 1

Tell me about your FP&A project.

You should be able to explain it in 60–90 seconds.

Question 2

Why did you use SQL?

Answer conceptually:

SQL was used as the structured financial data layer. I used it to organize, query, aggregate and validate financial information before using the results in the FP&A model.

Question 3

Why did you use Excel?

Excel was used for the actual FP&A modeling process, including assumptions, budgeting, forecasting, P&L analysis, Budget vs Actual, variance analysis and scenario modeling.

Question 4

Why did you use Power BI?

Power BI was used to convert the financial results into management reporting, allowing performance to be analyzed by time, business unit, product, region and department.

126. INTERVIEW QUESTION — FORECAST

How did you build the forecast?

Answer:

I used a YTD run-rate approach. I compared year-to-date actual performance with the corresponding budget, applied the resulting performance ratio to the remaining budget period, and combined the forecast remainder with YTD actuals to produce the full-year outlook.

Do not memorize the wording.

Understand the logic.

127. INTERVIEW QUESTION — VARIANCE

How did you analyze variance?

Answer:

I calculated both absolute and percentage variance and then analyzed the variance by business unit, region, product and department to identify the underlying drivers rather than stopping at the total variance.

128. INTERVIEW QUESTION — DATA QUALITY

How did you make sure your numbers were correct?

Answer:

I built reconciliation checks between the SQL data, Excel model and Power BI reporting outputs. I also included model controls for revenue, COGS, gross profit, operating expenses and EBIT to ensure the financial statements tied together.

129. INTERVIEW QUESTION — LIMITATIONS

Is this real company data?

Correct answer:

No. It is a synthetic dataset created specifically for the portfolio project. I designed the business model and financial logic to simulate a realistic technology-company FP&A environment. The purpose was to demonstrate the end-to-end analytical workflow without using confidential or fabricated claims about a real company.

This answer is important.

Never imply that MadeUP is a real company.

130. THE RESUME CONNECTION

Once the project is complete, your resume can have:

Financial Model (XL)

MadeUP — FP&A Budget & Forecast Model

Power BI Dashboards

MadeUP — FP&A Management Reporting Dashboard

Data Querying (SQL)

MadeUP — Financial Data Analysis

These are three components of one project.

That is much stronger than inventing three unrelated projects.

131. WHAT "ADVANCED EXCEL" MEANS AFTER THIS PROJECT

You should not think:

"I know a lot of Excel formulas."

Instead:

"I can use Excel to build a financial planning model."

Your evidence will include:

- driver-based budgeting
- forecasting
- variance analysis
- scenario modeling

- P&L
- financial KPIs
- reconciliation
- PivotTables
- XLOOKUP
- SUMIFS
- IF/IFS
- structured tables
- management reporting

That is a defensible interpretation of advanced Excel for an entry-level FP&A role.

132. WHAT "SQL" MEANS AFTER THIS PROJECT

You should be able to say:

"I have hands-on SQL experience working with relational financial data."

You will have demonstrated:

- database creation
 - tables
 - relationships
 - joins
 - aggregations
 - CTEs
 - filtering
 - financial calculations
 - trend analysis
 - validation
 - exporting analytical datasets
-

133. WHAT "POWER BI" MEANS AFTER THIS PROJECT

You should be able to say:

"I have hands-on Power BI experience building financial management dashboards."

You will have demonstrated:

- data import

- Power Query
 - data modeling
 - relationships
 - DAX measures
 - KPI design
 - financial visualization
 - variance analysis
 - interactive management reporting
-

134. LEARNING LOG — WHAT TO RECORD

At the end of each major phase, add an entry.

Example:

```
## SQL – Completed

### What I Built
Created the MadeUP_FPA SQL Server database and financial tables.

### What I Learned
- Difference between fact and dimension tables
- Primary keys
- Foreign keys
- JOIN
- GROUP BY
- Aggregate functions

### What I Achieved
I can now query financial data by month, product, business unit and region.

### Interview Takeaway
SQL can be used by FP&A analysts to retrieve, validate and analyze financial information.
```

Do the same for Excel and Power BI.

135. PROJECT COMPLETION SCORECARD

Use this as your final checklist.

Business Understanding

- ☐ MadeUP business model documented
- ☐ Products documented
- ☐ Revenue models documented
- ☐ Regions documented
- ☐ Departments documented
- ☐ Cost structure documented
- ☐ KPIs documented

SQL

- ☐ SQL Server installed
- ☐ SSMS installed
- ☐ Database created
- ☐ Dimension tables created
- ☐ Fact tables created
- ☐ Synthetic data generated
- ☐ Queries written
- ☐ JOINS used
- ☐ Aggregations used
- ☐ CTE used
- ☐ Financial analysis completed
- ☐ Validation queries completed
- ☐ CSV exports created

Excel

- ☐ Workbook created
- ☐ Actual data loaded
- ☐ Assumptions created
- ☐ Budget built
- ☐ P&L built
- ☐ Budget vs Actual built
- ☐ Variance analysis built
- ☐ Forecast built
- ☐ Scenario analysis built
- ☐ Management summary built
- ☐ Model checks built
- ☐ MODEL PASS achieved

Power BI

- ☐ Power BI Desktop installed
- ☐ Data imported
- ☐ Power Query used
- ☐ Relationships created

- ☐ DAX measures created
- ☐ Executive dashboard built
- ☐ Revenue page built
- ☐ Budget vs Actual page built
- ☐ Forecast page built
- ☐ Opex page built
- ☐ Variance driver page built
- ☐ Dashboard validated

Reconciliation

- ☐ SQL = Excel
- ☐ Excel = Power BI
- ☐ Revenue reconciles
- ☐ COGS reconciles
- ☐ Gross Profit reconciles
- ☐ Opex reconciles
- ☐ EBIT reconciles

Documentation

- ☐ README
- ☐ Business Model
- ☐ KPI Definitions
- ☐ Data Dictionary
- ☐ Methodology
- ☐ Reconciliation
- ☐ Management Commentary
- ☐ Limitations
- ☐ Screenshots
- ☐ Project Log

GitHub

- ☐ Repository public
- ☐ Files organized
- ☐ README polished
- ☐ SQL files uploaded
- ☐ Excel workbook uploaded
- ☐ Power BI file uploaded
- ☐ Screenshots uploaded
- ☐ Documentation uploaded
- ☐ No passwords/API keys/secrets
- ☐ Repository opens cleanly

136. DEFINITION OF 100% COMPLETE

The project is **100% complete** only when all of the following are true:

```
graph TD; A[Business Model] --> B[Synthetic Data]; B --> C[SQL Database]; C --> D[SQL Analysis]; D --> E[Excel Budget]; E --> F[Excel Forecast]; F --> G[Variance Analysis]; G --> H[Scenario Analysis]; H --> I[Power BI Dashboard]; I --> J[Reconciliation]; J --> K[Management Insights]; K --> L[Documentation]; L --> M[GitHub]; M --> N[Interview Explanation];
```

Business Model
↓
Synthetic Data
↓
SQL Database
↓
SQL Analysis
↓
Excel Budget
↓
Excel Forecast
↓
Variance Analysis
↓
Scenario Analysis
↓
Power BI Dashboard
↓
Reconciliation
↓
Management Insights
↓
Documentation
↓
GitHub
↓
Interview Explanation

If one major component is missing, the project is not finished.

137. EXPECTED TIME

Because you are learning SQL and Power BI while building, do not expect this to be a two-hour project.

A realistic fast-track target is approximately:

Phase	Approx. Time
Setup	1 hour
Business design	1 hour
SQL learning/build	4–6 hours
Excel model	5–8 hours
Power BI	4–6 hours
Reconciliation	1–2 hours
Documentation/GitHub	2–3 hours
Final review	1–2 hours

Core project

Approximately:

18–28 focused hours

The purpose is not to spend months building it.

The purpose is to build something **small enough to finish but sophisticated enough to defend**.

138. DO NOT ADD THESE THINGS YET

To keep the project fast, do not add:

- Python
- Tableau
- VBA
- machine learning
- advanced statistical forecasting
- complicated database administration
- hundreds of products
- dozens of regions
- millions of transactions
- real-company financial data
- web scraping
- APIs
- cloud deployment
- Power BI Service administration

Those can be future projects.

They are not required for this one.

139. OPTIONAL PHASE 2 — ONLY AFTER 100%

After the core project is complete, you may add:

Advanced SQL

- window functions
- more CTEs
- views
- stored procedures

Advanced Excel

- more sophisticated forecasting
- sensitivity tables
- more advanced model controls

Power BI

- more sophisticated time intelligence
- drill-through
- bookmarks
- tooltips
- advanced DAX

FP&A

- working capital
- cash flow forecasting
- long-range planning
- rolling forecast automation
- business cases

But do not touch these until the core project works.

140. THE MOST IMPORTANT RULE OF THIS MANUAL

At every point, ask:

What FP&A problem am I solving?

Not:

"What cool Excel function can I use?"

Not:

"What complicated SQL query can I write?"

Not:

"What fancy Power BI visual can I create?"

Instead:

"What financial question does management need answered?"

Then choose the tool.

141. THE TOOL MENTAL MODEL

Memorize this:

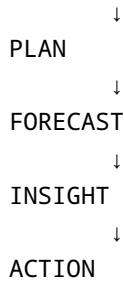
```
SQL
  "What happened?"
    ↓
DATA

EXCEL
  "What should happen?"
    ↓
PLANNING / MODELING

POWER BI
  "How do we communicate what happened?"
    ↓
REPORTING
```

And FP&A sits above all three:

```
BUSINESS QUESTION
  ↓
FINANCIAL DATA
  ↓
ANALYSIS
```

142. THE FINAL STORY OF THE PROJECT

When finished, the project should tell this story:

MadeUP is a fictional global technology company with multiple product and service businesses operating across four regions. I built a synthetic financial environment representing its historical results, annual planning process, current-year performance and forecast.

I used SQL to structure, query, aggregate and validate the company's financial data.

I used Microsoft Excel to build the FP&A model, including assumptions, budgeting, P&L analysis, Budget vs Actual, variance analysis, forecasting and scenario planning.

I then used Power BI to convert the financial results into management dashboards showing KPIs, performance trends, budget variances, forecast outlook and variance drivers.

Finally, I reconciled the outputs across the three tools and documented the methodology, assumptions, controls and management recommendations in a public GitHub repository.

That is the story you are building.

143. WHAT YOU ARE REALLY BUILDING

You are **not** building three software demonstrations.

You are building:

An FP&A analyst's workflow.

SQL is one part.

Excel is one part.

Power BI is one part.

The actual project is:

```
FINANCIAL DATA
+
FINANCIAL MODEL
+
FINANCIAL ANALYSIS
+
MANAGEMENT REPORTING
+
DECISION SUPPORT
```

That is why this project will be relevant to the J&J FP&A role.

144. FIRST ACTION — START HERE

Do not start with Excel.

Do not start with Power BI.

Do not start with SQL.

Start with:

STEP 1

Create:

```
MadeUP-FPA-Project
```

on your computer.

STEP 2

Create the folder structure from Section 19.

STEP 3

Create the public GitHub repository:

MadeUP-FPA-Project

STEP 4

Create:

08_Project_Log/PROJECT_PROGRESS.md

STEP 5

Write down:

Project:
MadeUP – Integrated FP&A Analytics Project

Industry:
Technology

Purpose:
Demonstrate hands-on FP&A capability using SQL,
Microsoft Excel, and Power BI.

Data:
Synthetic

Reporting Date:
June 30, 2026

STEP 6

Install:

- SQL Server 2025 Express
- SQL Server Management Studio
- Power BI Desktop

STEP 7

Stop.

Do not build the database yet.

Your next working session should begin with:

Phase 1 — Business Requirements

and then move sequentially through this manual.

145. FINAL PROJECT PRINCIPLE

The project should be:

Synthetic.

Authentic.

Small enough to finish.

Complex enough to demonstrate skill.

Financially logical.

Technically defensible.

Professionally documented.

Publicly viewable.

Interview-ready.

And most importantly:

Every claim on your resume should have something inside this project that proves it.

That is the standard for MadeUP.

OFFICIAL TOOL REFERENCES

- [Power BI Desktop — Microsoft Learn](#)
- [SQL Server 2025 editions and capabilities — Microsoft Learn](#)
- [SQL Server Management Studio — Microsoft Learn](#)
- [Excel for the web — Microsoft Support](#)

- [Excel PivotTables — Microsoft Support](#)
- [Excel SUMIFS — Microsoft Support](#)
- [GitHub repository quickstart — GitHub Docs](#)
- [GitHub file uploads — GitHub Docs](#)

146. END STATE

When you reach the end of this manual, your resume will not simply say:

Advanced Microsoft Excel, Power BI, & SQL

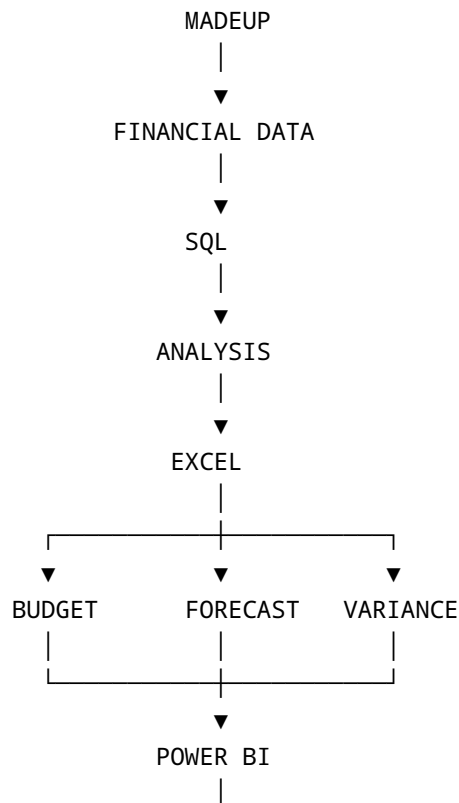
You will have a public project behind that statement.

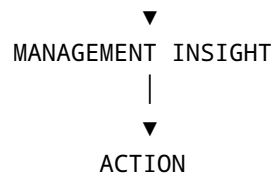
And when a recruiter asks:

"Can you tell me about your experience using Excel, SQL, and Power BI?"

you won't have to give them three unrelated examples.

You will be able to show them **one complete financial workflow**:





That is the project.

That is the portfolio evidence.

That is what you will build.